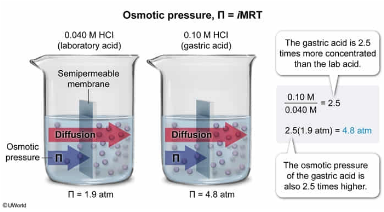
A 0.040 M HCl solution is measured in the laboratory and has an osmotic pressure of 1.9 atm. What is the estimated osmotic pressure of gastric acid (approximated as 0.10 M HCl) under the same conditions?

- A. 0.19 atm [3%]
- B. 0.40 atm [10%]
- C. 2.4 atm [13%]
- ✔ D. 4.8 atm [72%]

Correct
71% answered correctly

Explanation:



The diffusion of solvent across a **semipermeable membrane** from a solution of lower concentration into a solution of higher concentration during osmosis produces an **osmotic pressure** Π against the membrane, which is calculated by the relationship:

$$\Pi = i\,\mathrm{MRT}$$

Although the osmotic pressure of the gastric acid relative to the laboratory HCl can be calculated by this relationship, it is not necessary to do so here. As shown by the equation, osmotic pressure is **proportional** to the **concentration** of the osmotically active solute. Accordingly, because *i*RT is the same for both samples, the calculation can be performed as a proportion relating the gastric acid to the laboratory acid as follows:

$$\frac{\Pi_{\text{gastric acid}}}{\Pi_{\text{lab acid}}} = \frac{\mathrm{M}_{\text{gastric acid}}}{\mathrm{M}_{\text{lab acid}}}$$

$$\frac{\Pi_{\text{gastric acid}}}{1.9\ \text{atm}} = \frac{0.10\ \mathrm{M}}{0.040\ \mathrm{M}} =$$

$$\Pi_{\text{gastric acid}} = 2.5 \quad ($$

Because the gastric acid concentration is 2.5 times greater than that of the laboratory acid, the osmotic pressure of the gastric acid is also 2.5 times greater than that of the laboratory acid, giving the gastric acid an osmotic pressure of 4.8 atm.

(Choice A) Multiplying the concentration of the gastric acid by the osmotic pressure of the laboratory acid yields a value of 0.19 with incorrect units of atm·M. This value must also be divided by the laboratory acid concentration.

(Choice B) If the temperature of 24 °C is not converted to the Kelvin scale (24 °C + 273 = 297 K), a calculated value of 0.40 atm·°C/K (with incorrect units) is obtained.

(Choice C) Using a value of *i* = 1 does not account for the dissociation of HCl into two ions (H^+ and Cl^-). This yields a calculated result of 2.4 atm, which is only half of the correct value.

Educational objective:
The diffusion of solvent across a semipermeable membrane during osmosis produces an osmotic pressure against the membrane, which is directly proportional to the concentration of the osmotically active solute in solution.

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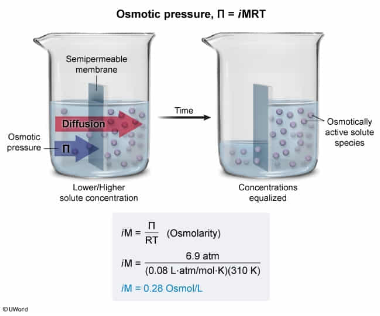
Osmolarity, the number of moles of osmotically active species (osmoles) per liter of solution (Osmol/L), equals the molar solute concentration M multiplied by the van 't Hoff factor i. Osmolarity is calculated by dividing osmotic pressure Π by RT, where R is the gas constant and T is the absolute temperature.

If the osmotic pressure of blood plasma is 6.9 atm at 37 °C, what is its osmolarity? (Note: Assume R = 0.08 L·atm/mol·K, and i = 2.)

- A. 0.14 Osmol/L [15%]
- ✔ B. 0.28 Osmol/L [63%]
- C. 1.2 Osmol/L [10%]
- D. 2.3 Osmol/L [10%]

Correct
63% answered correctly

Explanation:



The diffusion of solvent across a **semipermeable membrane** from a solution of lower concentration to a solution of higher concentration during osmosis produces an **osmotic pressure** against the membrane. Dissolved ions or molecules that cannot cross the membrane are **osmotically active** because they cause a concentration difference that induces the **solvent diffusion**. Accordingly, the osmotic pressure is proportional to the molar concentration of osmotically active solute species (the **osmolarity**), which is calculated by i·M.

The concentration (molarity, M) of the bulk solute is not necessarily equivalent to the osmolarity because some solutes can dissociate into ions. For example, NaCl dissociates into two ions (Na^+ and Cl^-), which causes its osmolarity to be twice the NaCl concentration. To account for this, the **van 't Hoff factor** i is used, which is a coefficient related to the number of moles of osmotically active ions or molecules (osmoles) that a solute molecule will form when dissolved. In complex solutions with mixtures of different solutes, determining i can become complicated. Consequently, i may be estimated or the measured osmolarity may be evaluated directly instead.

If the osmotic pressure of blood plasma is 6.9 atm at physiological temperature (37 °C), the osmolarity of the plasma can be calculated as:

$$i \qquad M \qquad =$$

(Choice A) Solving for molarity (M) instead of osmolarity (i·M) results in a value of 0.14 mol/L, which is the concentration of the nondissociated solutes rather than the concentration of all osmotically active species.

(Choice C) If the temperature is not converted from Celsius to Kelvin and the equation is also solved for molarity (M) instead of osmolarity (i·M), a value of 1.2 mol·K/°C·L is obtained (incorrect units).

(Choice D) Using the temperature in Celsius instead of Kelvin gives a value of 2.3 mol·K/°C·L (incorrect units).

Educational objective:
Solvent diffusion across the semipermeable membrane during osmosis produces an osmotic pressure against the membrane proportional to the osmolarity (concentration of osmotically active species that cannot cross the membrane) rather than to the solute molarity due to the dissociation of some solutes. Solute concentration and osmolarity are related by the van 't Hoff factor.

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